2-FIBRE PRODUCTS OF ALGEBRAIC STACKS OVER A STACK WITH SPACE-REPRESENTABLE DIAGONAL ARE ALGEBRAIC

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Editorial note. No omission markers in-span; the Isom square commutativity is read off the explicit object descriptions (mechanical). The two property-transfer inputs are not carded ('Proof omitted.' each) and are declared external.

Allowed standard prerequisites (editorial).

- Stack closure under 2-fibre products. Stacks, lemma 2-product-stacks-in-groupoids (cross-chapter, named). Condition (1).
- Diagonal criterion. lemma-representable-diagonal (carded L26680). Isom sheaves of algebraic-stack inputs are algebraic spaces.
- Base-change property transfer. lemma-base-change-representable-transformations-property (same chapter, NOT carded, declared external). Properties transfer along base change.
- Composition property transfer. lemma-composition-representable-transformations-property (same chapter, NOT carded, declared external). Properties transfer along composition.

Source reference. Original reference 0040: focused-seeds/geometry/S9-alg-2-fibre-product-general/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 026O: sources/stacks-algebraic/pinned-original-source.tex, source lines 1506–1506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045C: sources/stacks-algebraic/pinned-original-source.tex, source lines 1180–1180. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045B: sources/stacks-algebraic/pinned-original-source.tex, source lines 1159–1159. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 045G: sources/stacks-algebraic/pinned-original-source.tex, source lines 1337–1337. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02ZL: focused-seeds/geometry/S9-alg-2-fibre-product-general/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{Z}$ be a stack in groupoids over $(Sch/S)_{fppf}$ whose diagonal is representable by algebraic spaces. Let $\mathcal{X}$, $\mathcal{Y}$ be algebraic stacks over S. Let $f : \mathcal{X} \to \mathcal{Z}$, $g : \mathcal{Y} \to \mathcal{Z}$ be 1-morphisms of stacks in groupoids. Then the 2-fibre product $\mathcal{X} \times_{f,\mathcal{Z},g} \mathcal{Y}$ is an algebraic stack.*

Proof. We have to check conditions (1), (2), and (3) of Definition 026O. The first condition follows from Stacks, Lemma 02ZL.

The second condition we have to check is that the *Isom*-sheaves are representable by algebraic spaces. To do this, suppose that T is a scheme over S, and u, v are objects of $(\mathcal{X} \times_{f,\mathcal{Z},g} \mathcal{Y})_T$. By our construction of 2-fibre products (which goes all the way back to Categories, Lemma 0040) we may write $u = (x, y, \alpha)$ and $v = (x', y', \alpha')$. Here $\alpha : f(x) \to g(y)$ and similarly for α'. Then it is clear that

$$\begin{array}{ccc} Isom(u, v) & \longrightarrow & Isom(y, y') \\ \downarrow & & \downarrow{\scriptstyle \phi \mapsto g(\phi) \circ \alpha} \\ Isom(x, x') & \xrightarrow{\psi \mapsto \alpha' \circ f(\psi)} & Isom(f(x), g(y')) \end{array}$$

is a cartesian diagram of sheaves on $(Sch/T)_{fppf}$. Since by assumption the sheaves $Isom(y, y')$, $Isom(x, x')$, $Isom(f(x), g(y'))$ are algebraic spaces (see Lemma 045G) we see that $Isom(u, v)$ is an algebraic space.

Let $U, V \in \mathrm{Ob}((Sch/S)_{fppf})$, and let x, y be surjective smooth morphisms $x : (Sch/U)_{fppf} \to \mathcal{X}$, $y : (Sch/V)_{fppf} \to \mathcal{Y}$. Consider the morphism

$$(Sch/U)_{fppf} \times_{f \circ x, \mathcal{Z}, g \circ y} (Sch/V)_{fppf} \longrightarrow \mathcal{X} \times_{f,\mathcal{Z},g} \mathcal{Y}.$$

As the diagonal of $\mathcal{Z}$ is representable by algebraic spaces the source of this arrow is representable by an algebraic space F, see Lemma 045G. Moreover, the morphism is the composition of base changes of x and y, hence surjective and smooth, see Lemmas 045C and 045B. Choosing a scheme W and a surjective étale morphism $W \to F$ we see that the composition of the displayed 1-morphism with the corresponding 1-morphism

$$(Sch/W)_{fppf} \longrightarrow (Sch/U)_{fppf} \times_{f \circ x, \mathcal{Z}, g \circ y} (Sch/V)_{fppf}$$

is surjective and smooth which proves the last condition. □